

Ficha 2

$$1. f((a,b),c) = (a \wedge b) \oplus c \rightarrow \text{xor}((a \wedge b), c) \stackrel{\text{and}}{=} \text{xor}(\text{and}(a,b), c)$$

$$\stackrel{74}{=} \text{xor}(\text{and}(a,b), \text{id } c) \stackrel{78}{=} \text{xor}(\text{and} \times \text{id}((a,b), c)) \stackrel{73 \text{ e } 72}{=} \text{xor}(\text{and} \times \text{id})$$

$$f = \text{xor} \cdot (\text{and} \times \text{id})$$

$$2. g = \langle \pi_1, f \rangle$$

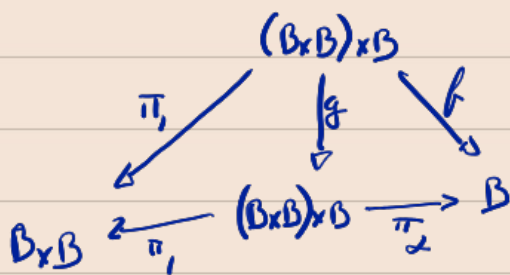
$$\text{Tipo } f : (B \times B) \times B \rightarrow B$$

$$\text{Tipo } \pi_1 : A \times C \rightarrow A$$

$$\text{Se } A = B \times B$$

$$C = B$$

$$\therefore \pi_1 : (B \times B) \times B \rightarrow B \times B$$



Como $g = \langle \pi_1, f \rangle$

$$\therefore g : (B \times B) \times B \rightarrow (B \times B) \times B$$

resultado de π_1 (pointing to π_1)
resultado de f (pointing to f)
ambos recebem isto (pointing to the domain $(B \times B) \times B$)
 $\langle \rangle$ (pointing to the codomain $(B \times B) \times B$)

$$4. \kappa = \langle f, g \rangle \stackrel{6}{=} \begin{cases} \pi_1 \cdot \kappa = f \\ \pi_2 \cdot \kappa = g \end{cases}$$

$$\kappa = \text{id} \Rightarrow \text{id} = \langle f, g \rangle \stackrel{6}{=} \begin{cases} \pi_1 = f \\ \pi_2 = g \end{cases} \Leftrightarrow \text{id} = \langle \pi_1, \pi_2 \rangle \stackrel{8}{=} \langle \pi_1, \pi_2 \rangle = \text{id}_{A \times B}$$

$$5. \begin{cases} \pi_1 \cdot (\langle h, \kappa \rangle \cdot f) = h \cdot f \\ \pi_2 \cdot (\langle h, \kappa \rangle \cdot f) = \kappa \cdot f \end{cases} \stackrel{2}{=} \begin{cases} (\pi_1 \cdot \langle h, \kappa \rangle) \cdot f = h \cdot f \\ (\pi_2 \cdot \langle h, \kappa \rangle) \cdot f = \kappa \cdot f \end{cases} \stackrel{7}{=} \begin{cases} h \cdot f = h \cdot f \\ \kappa \cdot f = \kappa \cdot f \end{cases} \text{ c.q.m.}$$

$$6. \text{dup} \cdot f = \langle f, f \rangle \quad \text{dup } x = (x, x) \stackrel{74}{=} \text{dup } x = (\text{id } x, \text{id } x) \stackrel{74, 72}{=} \text{dup} = \langle \text{id}, \text{id} \rangle$$

$$\langle \text{id}, \text{id} \rangle \cdot f = \langle f, f \rangle \stackrel{6}{=} \begin{cases} \pi_1 \cdot \langle \text{id}, \text{id} \rangle \cdot f = f \\ \pi_2 \cdot \langle \text{id}, \text{id} \rangle \cdot f = f \end{cases} \stackrel{27}{=} \begin{cases} \text{id} \cdot f = f \\ \text{id} \cdot f = f \end{cases} \stackrel{1}{=} \begin{cases} f = f \\ f = f \end{cases} \text{ c.q.m.}$$

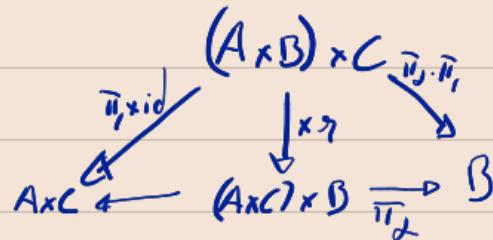
$$7. \gamma\gamma = \langle \pi, xid, \pi_2 \cdot \pi, \rangle$$

$$x\gamma \cdot \langle \langle f, g \rangle, h \rangle = \langle \langle f, h \rangle, g \rangle$$

$$\langle \pi, xid, \pi_2 \cdot \pi, \rangle \cdot \langle \langle f, g \rangle, h \rangle \stackrel{9}{=} \langle \langle \pi, xid \rangle \cdot \langle f, g \rangle, (\pi_2 \cdot \pi) \cdot \langle f, g \rangle, h \rangle \stackrel{11,2}{=} \langle \langle \pi, \langle f, g \rangle, id \cdot h \rangle, \pi_2 \cdot \langle \pi, \langle f, g \rangle, h \rangle \rangle$$

$$\stackrel{7,7,7}{=} \langle \langle f, h \rangle, g \rangle \text{ c.q.m.}$$

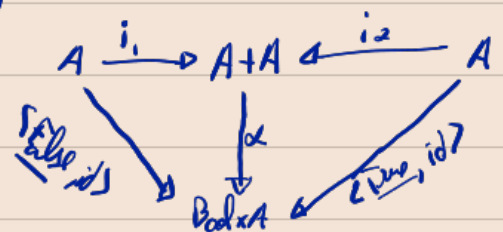
$$x\gamma: (A \times B) \times C \rightarrow (A \times C) \times B$$



$$8. \underline{(b, a)} = \langle \underline{b}, \underline{a} \rangle \stackrel{6,4}{=} \left\{ \begin{array}{l} \underline{\pi_1} \cdot \underline{(b, a)} = \underline{b} \\ \underline{\pi_2} \cdot \underline{(b, a)} = \underline{a} \end{array} \right\} \stackrel{7,9}{=} \left\{ \begin{array}{l} \underline{b} = \underline{b} \\ \underline{a} = \underline{a} \end{array} \right\} \text{ c.q.m.}$$

$$9. \alpha = [\langle \underline{\text{False}}, id \rangle, \langle \underline{\text{True}}, id \rangle]$$

$$\left. \begin{array}{l} \text{False} : A \rightarrow \text{Bool} \\ id : * \rightarrow * \\ \text{True} : B \rightarrow \text{Bool} \end{array} \right\} \left\{ \begin{array}{l} \langle \underline{\text{False}}, id \rangle : A \rightarrow \text{Bool} \times A \\ \langle \underline{\text{True}}, id \rangle : B \rightarrow \text{Bool} \times B \end{array} \right\} \alpha : A + A \rightarrow \text{Bool} \times A$$



Ficha 3

$$1. \text{assoc} = \langle id \times \pi_1, \pi_2 \cdot \pi_2 \rangle$$

$$\begin{aligned}
 \text{assoc} \cdot \text{assoc} = \text{id} &\equiv \langle \text{id} \times \pi_1, \pi_2 \cdot \pi_2 \rangle \cdot \text{assoc} \stackrel{9}{=} \langle (\text{id} \times \pi_1) \cdot \text{assoc}, (\pi_2 \cdot \pi_2) \cdot \text{assoc} \rangle \stackrel{6,1}{=} \left\{ \begin{array}{l} (\text{id} \times \pi_1) \cdot \text{assoc} = \pi_1 \\ \pi_2 \cdot \pi_2 \cdot \text{assoc} = \pi_2 \end{array} \right. \\
 \text{id}_1 &\equiv \left\{ \begin{array}{l} \langle \pi_1, \pi_1 \cdot \pi_2 \rangle \cdot \text{assoc} = \pi_1 \\ \pi_2 \cdot \pi_2 \cdot \text{assoc} = \pi_2 \end{array} \right. \stackrel{9,6}{=} \left\{ \begin{array}{l} \pi_1 \cdot \pi_1 \cdot \pi_2 = \pi_1 \cdot \pi_1 \\ \pi_1 \cdot \pi_2 \cdot \pi_2 = \pi_2 \cdot \pi_1 \end{array} \right. \stackrel{6}{=} \left\{ \begin{array}{l} \pi_1 \cdot \pi_2 \cdot \pi_2 = \pi_2 \cdot \pi_1 \\ \pi_2 \cdot \pi_2 \cdot \pi_2 = \pi_2 \end{array} \right. \\
 &\stackrel{6}{=} \left\{ \begin{array}{l} \pi_2 \cdot \pi_2 = \langle \pi_2 \cdot \pi_1, \pi_2 \rangle \\ \text{assoc} = \langle (\pi_1 \cdot \pi_1), \langle \pi_2 \cdot \pi_1, \pi_2 \rangle \rangle \end{array} \right. \\
 &\quad \downarrow \\
 &\quad \langle \pi_2 \cdot \pi_1, \text{id} \cdot \pi_2 \rangle = \pi_2 \times \text{id}
 \end{aligned}$$

$$\begin{aligned}
 \langle \text{id} \times \pi_1, \pi_2 \cdot \pi_2 \rangle &: A \times (B \times C) \rightarrow (A \times B) \times C \\
 \downarrow &\quad \searrow \\
 A \times (B \times C) &\rightarrow A \times B \quad A \times (B \times C) \rightarrow C
 \end{aligned}$$

$$3. \kappa = [f, g] \equiv \begin{cases} \kappa \cdot i_1 = f \\ \kappa \cdot i_2 = g \end{cases}$$

$$[\kappa, \kappa] = \kappa \stackrel{17}{=} \begin{cases} \kappa \cdot i_1 = \kappa \\ \kappa \cdot i_2 = \kappa \end{cases} \stackrel{3}{=} \begin{cases} \kappa = \kappa \\ \kappa = \kappa \end{cases} \text{ c.q.m.}$$

$$5. \alpha = [\langle \underline{\text{False}}, \text{id} \rangle, \langle \underline{\text{True}}, \text{id} \rangle] \stackrel{17}{=} \begin{cases} \alpha \cdot i_1 = \langle \underline{\text{False}}, \text{id} \rangle \\ \alpha \cdot i_2 = \langle \underline{\text{True}}, \text{id} \rangle \end{cases} \stackrel{7,7,23}{=} \begin{cases} \alpha(i_1, a) = \langle \underline{\text{False}}, \text{id} \rangle a \\ \alpha(i_2, a) = \langle \underline{\text{True}}, \text{id} \rangle a \end{cases}$$

$$\stackrel{7,7}{=} \begin{cases} \alpha(i_1, a) = (\underline{\text{False}} a, \text{id} a) \\ \alpha(i_2, a) = (\underline{\text{True}} a, \text{id} a) \end{cases} \stackrel{4,7,4}{=} \begin{cases} \alpha(i_1, a) = (\underline{\text{False}}, a) \\ \alpha(i_2, a) = (\underline{\text{True}}, a) \end{cases} \text{ c.q.m.}$$

$$6. \text{fac } 0 = 1 \\
 \text{fac}(m+1) = (m+1) * \text{fac } m \quad \Leftrightarrow \text{fac} \cdot [0, \text{succ}] = [1, \text{mult} \cdot \langle \text{succ}, \text{fac} \rangle]$$

$$\text{succ } m = m + 1$$

$$\text{mul}(a, b) = a * b$$

$$\text{fac} \cdot [0, \text{succ}] = [1, \text{mul} \cdot \langle \text{succ}, \text{fac} \rangle] = [\text{fac} \cdot 0, \text{fac} \cdot \text{succ}] = [1, \text{mul} \cdot \langle \text{succ}, \text{fac} \rangle]$$

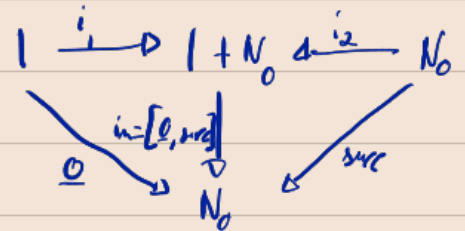
$$\stackrel{27}{=} \begin{cases} \text{fac} \cdot 0 = 1 \\ \text{fac} \cdot \text{succ} = \text{mul} \cdot \langle \text{succ}, \text{fac} \rangle \end{cases} \stackrel{72}{=} \begin{cases} (\text{fac } 0) a = 1 a \\ (\text{fac } \text{succ}) a = \text{mul} \langle \text{succ}, \text{fac} \rangle a \end{cases}$$

$$\stackrel{73, 75, 77}{=} \begin{cases} \text{fac } 0 = 1 \\ \text{fac}(a+1) = \text{mul}(a+1, \text{fac } a) \end{cases} \stackrel{\text{mul}}{=} \begin{cases} \text{fac } 0 = 1 \\ \text{fac}(a+1) = (a+1) \times \text{fac } a \end{cases} \text{ c.g.m.}$$

$$7. \text{ in} = [0, \text{succ}]$$

$$\text{in} \cdot \text{out} = \text{id}$$

$$\begin{cases} \text{out } 0 = i_1() \\ \text{out}(a+1) = i_2 a \end{cases}$$



$$\text{in} \cdot \text{out} = \text{id} \stackrel{\text{in}}{=} [0, \text{succ}] \cdot \text{out} = \text{id} \stackrel{20}{=} [0 \cdot \text{out}, \text{succ} \cdot \text{out}] = \text{id} \stackrel{17}{=} \begin{cases} 0 \cdot \text{out} = i_1 \cdot i_1 \\ \text{succ} \cdot \text{out} = i_2 \cdot i_2 \end{cases}$$

$$\stackrel{4}{=} \begin{cases} \text{out } 0 = i_1 \\ \text{out } \text{succ} = i_2 \end{cases} \stackrel{72}{=} \begin{cases} \text{out } 0() = i_1() \\ \text{out } \text{succ } a = i_2 a \end{cases} \stackrel{\text{succ}}{=} \begin{cases} \text{out } 0 = i_1() \\ \text{out}(a+1) = i_2 a \end{cases} \text{ c.g.m.}$$

$$8. \text{ fac} = [1, \text{mul}] \cdot (\text{id} + \langle \text{succ}, \text{fac} \rangle) \cdot \text{out} \stackrel{22}{=} [1 \cdot \text{id}, \text{mul} \cdot \langle \text{succ}, \text{fac} \rangle] \cdot \text{out} \stackrel{19}{=} [1, \langle \text{mul} \cdot \text{succ}, \text{mul} \cdot \text{fac} \rangle] \cdot \text{out}$$

$$= [1,$$

Ficha 4

$$1. \text{ coarsel} = [\text{id} + i_1, i_2 \cdot i_2]$$

$$\text{coarsel} \cdot [\text{id} + i_1, i_2 \cdot i_2] = \text{id} \stackrel{20}{=} [\text{coarsel} \cdot (\text{id} + i_1), \text{coarsel} \cdot i_2 \cdot i_2] = \text{id} \stackrel{17}{=} \begin{cases} \text{coarsel} \cdot (\text{id} + i_1) = i_1 \\ \text{coarsel} \cdot i_2 \cdot i_2 = i_2 \end{cases}$$

$$\stackrel{21, 1}{=} \text{coarsel} \cdot [i_1, i_2 \cdot i_2] = i_1 \stackrel{20}{=} \text{coarsel} \cdot i_1, \text{coarsel} \cdot i_2 \cdot i_2 = i_2$$

$$\begin{aligned}
 & \text{coassoc} \cdot i_1 \cdot i_2 = i_2 \\
 & \text{coassoc} \cdot i_1 \cdot i_2 = i_2 \\
 17 \equiv & \begin{cases} \text{coassoc} \cdot i_1 = i_1 \cdot i_1 \\ \text{coassoc} \cdot i_2 \cdot i_1 = i_1 \cdot i_2 \\ \text{coassoc} \cdot i_2 \cdot i_2 = i_2 \end{cases} \quad 17 \equiv \begin{cases} \text{coassoc} \cdot i_1 = i_1 \cdot i_1 \\ \text{coassoc} \cdot i_2 = [i_1 \cdot i_2, i_2] \end{cases} \quad 17 \equiv \text{coassoc} = [i_1 \cdot i_1, [i_1 \cdot i_2, i_2]] \stackrel{1,21}{=} \text{coassoc} = [i_1 \cdot i_1, i_2 + id]
 \end{aligned}$$

2. $in = [Leaf, Fork]$ $out \cdot in = id \stackrel{F1}{=} out \cdot [Leaf, Fork] = id \stackrel{20}{=} [out \cdot Leaf, out \cdot Fork] = id \equiv$

$$\begin{aligned}
 & out(Leaf\ a) = i_1\ a \\
 & out(Fork(x, y)) = i_2(x, y) \\
 & \stackrel{1,17}{=} \begin{cases} out \cdot Leaf = i_1 \\ out \cdot Fork = i_2 \end{cases} \stackrel{21,23}{=} \begin{cases} out(Leaf\ a) = i_1\ a \\ out(Fork(x, y)) = i_2(x, y) \end{cases} \quad \text{c.q.m}
 \end{aligned}$$

3. $\alpha = (id + \pi_1) \cdot i_2 \cdot \pi_2$

$$\alpha : C + B \xrightarrow{id + \pi_1} C + (B \times D) \xrightarrow{i_2} B \times D \xrightarrow{\pi_2} A \times (B \times D)$$

$$\alpha : A \times (B \times D) \rightarrow C + B$$

4. $\alpha = \text{swap} \cdot (id \times \text{swap})$

$$\text{swap} : A \times B \rightarrow B \times A$$

$$id \times \text{swap} : A \times (B \times C) \rightarrow A \times (C \times B)$$

$$\alpha : A \times (B \times C) \xrightarrow{id \times \sim} A \times (C \times B) \xrightarrow{\sim} (C \times B) \times A$$

$$\begin{array}{ccc}
 A \times (B \times C) & \xrightarrow{\alpha} & (C \times B) \times A \\
 \downarrow & & \downarrow \\
 f \times (g \times h) & & (h \times g) \times f \\
 \downarrow & & \downarrow \\
 A' \times (B' \times C') & \xrightarrow{\alpha} & (C' \times B') \times A'
 \end{array}$$

$$\alpha \cdot (f \times (g \times h)) = ((h \times g) \times f) \cdot \alpha$$

5. $\text{join} = [id, id] : A + A \rightarrow A$

$$\text{dup} = \langle id, id \rangle : B \rightarrow B \times B$$

$$\alpha = \text{dup} \cdot \text{join} : A + A \rightarrow A \times A$$

→ caso fosse $\text{join} \cdot \text{dup}$, o tipo de saída do dup não iria ser igual ao tipo de entrada do join .

$$A + A \xrightarrow{\alpha} A \times A$$

$$\begin{array}{ccc} f + f & & f \times f \\ \downarrow & & \downarrow \\ A' + A' & \xrightarrow{\alpha} & A' \times A' \end{array}$$

$$\alpha \cdot (f + f) = (f \times f) \cdot \alpha$$

6. $\nabla \cdot i_1 = id$
 $\nabla \cdot i_2 = id$

$$f \cdot \nabla = \nabla \cdot (f + f) \stackrel{\nabla}{=} f \cdot [id, id] = [id, id] \cdot (f + f) \stackrel{20, 22}{=} [f \cdot id, f \cdot id] = [id \cdot f, id \cdot f] \stackrel{!}{=} [f \cdot f] = [f, f] \text{ c.q.m.}$$

$$\left\{ \begin{array}{l} \nabla \cdot i_1 = id \\ \nabla \cdot i_2 = id \end{array} \right. \stackrel{!}{=} \nabla = [id, id]$$

7. $(f + h) \cdot \alpha = \alpha \cdot (f + g \times h)$

$$\alpha: A + (B \times C) \rightarrow A + C$$

$$id + \pi_2$$

$$\begin{array}{ccc} A + (B \times C) & \xrightarrow{\alpha} & A + C \\ \downarrow & & \downarrow \\ f + (g \times h) & & f + h \\ \downarrow & & \downarrow \\ A' + (B' \times C') & \xrightarrow{\alpha} & A' + C' \end{array}$$

$$(f + h) \cdot \alpha = \alpha \cdot (f + g \times h) \stackrel{!}{=} (f + h) \cdot (id + \pi_2) = (id + \pi_2) \cdot (f + g \times h) \stackrel{!}{=}$$

$$\stackrel{25}{=} (f \cdot id) + (h \cdot \pi_2) = (id \cdot f) + (\pi_2 \cdot (g \times h)) \stackrel{1, 13}{=} f + (h \cdot \pi_2) = f + (h \cdot \pi_2) \text{ c.q.m.}$$

8. $\alpha \cdot g = h \stackrel{!}{=} g = \alpha^o \cdot h$
 $g \cdot \alpha = h \stackrel{!}{=} g = h \cdot \alpha^o$

$$h \cdot distr \cdot (g \times (id + \alpha)) = K \stackrel{!}{=}$$

$$\begin{array}{ccc} A \times (B + C) & \xrightarrow{distr} & (A \times B) + (A \times C) \\ \downarrow & & \downarrow \\ f \times (g + h) & & (f \times g) + (f \times h) \\ \downarrow & & \downarrow \\ A' \times (B' + C') & \xrightarrow{distr} & (A' \times B') + (A' \times C') \end{array}$$

$$distr \cdot (f \times (g + h)) = (f \times g) + (f \times h) \cdot distr$$

P. 6. $\stackrel{!}{=} h \cdot ((g \times id) + (g \times \alpha)) \cdot distr = K \stackrel{!}{=} h \cdot ((g \times id) + (g \times \alpha)) = K \cdot distr \text{ c.q.m.}$

9. $[<f, g>, <h, k>] = [<f, h>, <g, k>] \stackrel{6}{=} \left\{ \begin{array}{l} \pi_1 \cdot [<f, g>, <h, k>] = [f, h] \\ \pi_2 \cdot [<f, g>, <h, k>] = [g, k] \end{array} \right. \stackrel{20}{=} \left\{ \begin{array}{l} [\pi_1 \cdot <f, g>, \pi_1 \cdot <h, k>] = [f, h] \\ \pi_2 \cdot [<f, g>, <h, k>] = [g, k] \end{array} \right. \stackrel{!}{=}$

$$\begin{matrix} \vdash \\ \equiv \end{matrix} \begin{cases} [f, h] = [f, h] \\ [g, k] = [g, k] \end{cases} \text{ c.g.m.}$$

Ficha 5

$$\begin{aligned} 1. (p \rightarrow f, g) \cdot h &= (p \cdot h) \rightarrow (f \cdot h, g \cdot h) \stackrel{30}{=} ([f, g] \cdot p?) \cdot h = \underbrace{[(f \cdot h) \cdot (g \cdot h)] \cdot (p \cdot h)?}_{\stackrel{2}{=} [f, g] \cdot (p? \cdot h) \stackrel{29}{=} [f, g] \cdot (h+h) \cdot (p \cdot h)? \stackrel{2}{=} ([f, g] \cdot (h+h)) \cdot (p \cdot h)?} \cdot h) \cdot (p \cdot h)? \equiv \\ &\stackrel{12}{=} [(f \cdot h), (g \cdot h)] \cdot (p \cdot h)? \quad \leftarrow \text{c.g.m.} \end{aligned}$$

$$\begin{aligned} 2. \text{parallel } pfg &= \langle \langle f, g \rangle, p \rangle \\ \text{choose } = \bar{u}_2 &\rightarrow \bar{u}_1, \bar{u}_1, \bar{u}_2, \bar{u}_1 \end{aligned}$$

$$\text{choose} \cdot \text{parallel } pfg = p \rightarrow f, g \stackrel{\text{ch, pa}}{=} (\bar{u}_2 \rightarrow \bar{u}_1, \bar{u}_1, \bar{u}_2, \bar{u}_1) \cdot \langle \langle f, g \rangle, p \rangle \equiv$$

$$\stackrel{32}{=} (\bar{u}_2 \cdot \langle \langle f, g \rangle, p \rangle) \rightarrow ((\bar{u}_1, \bar{u}_1) \cdot \langle \langle f, g \rangle, p \rangle), (\bar{u}_2, \bar{u}_1) \cdot \langle \langle f, g \rangle, p \rangle) \equiv$$

$$\stackrel{2,7}{=} p \rightarrow ((\bar{u}_1, \bar{u}_1) \cdot \langle f, g \rangle), (\bar{u}_2, \bar{u}_1) \cdot \langle f, g \rangle) \stackrel{7}{=} p \rightarrow \langle f, g \rangle \quad \text{c.g.m.}$$

$$3. (F5) \langle (p \rightarrow f, h), (p \rightarrow g, i) \rangle = p \rightarrow \langle f, g \rangle, \langle h, i \rangle \equiv$$

$$\stackrel{30}{=} \langle ([f, h] \cdot p?), ([g, i] \cdot p?) \rangle \stackrel{9}{=} \langle [f, h], [g, i] \rangle \cdot p? \stackrel{28}{=} \langle \langle f, g \rangle, \langle h, i \rangle \rangle \cdot p? \stackrel{30}{=} p \rightarrow \langle f, g \rangle, \langle h, i \rangle \quad \text{c.g.m.}$$

$$(F6) \langle f, (p \rightarrow g, h) \rangle = p \rightarrow \langle f, g \rangle, \langle f, h \rangle \equiv$$

$$\stackrel{30}{=} \langle \langle f, g \rangle, \langle f, h \rangle \rangle \cdot p? \stackrel{28}{=} \langle [f, f], [g, h] \rangle \cdot p? \stackrel{9}{=} \langle [f, f] \cdot p?, [g, h] \cdot p? \rangle \equiv$$

$$\stackrel{30}{=} \langle p \rightarrow f, f, p \rightarrow g, h \rangle \stackrel{F2}{=} \langle f, p \rightarrow g, h \rangle \quad \text{c.g.m.}$$

$$(F7) \quad p \rightarrow (p \rightarrow a, b), (p \rightarrow c, d) = p \rightarrow a, d$$

$$\stackrel{30}{=} p \rightarrow ([a, b] \cdot p?), ([c, d] \cdot p?) \stackrel{30}{=} [[a, b] \cdot p?, [c, d] \cdot p?] \cdot p? \stackrel{42}{=} ([[a, b], [c, d]] \cdot (p? + p?)) \cdot p? \equiv$$

$$\stackrel{J, F4}{=} [[a, b], [c, d]] \cdot (i_1 + i_2) \cdot p? \stackrel{42}{=} [[a, b] \cdot i_1, [c, d] \cdot i_2] \cdot p? \stackrel{18, 30}{=} p \rightarrow a, d \quad c.q.m.$$

$$4. \quad ap.(\bar{f} \times id) = \bar{f} \stackrel{21, 73}{=} \text{uncurry } f$$

$$\text{uncurry } f \ a \ b = f(a, b)$$

$$ap.(\bar{f} \times id) = \bar{f} \stackrel{73, 73}{=} ap(\bar{f} \times id)(a, b) = f(a, b) \stackrel{78, 1}{=} ap(\bar{f} a, b) = f(a, b) \equiv$$

$$\stackrel{84}{=} \bar{f} \ a \ b = f(a, b) \stackrel{85}{=} f(a, b) = f(a, b) \quad c.q.m.$$

$$5. \quad f := \text{uncurry } g = \hat{g}$$

$$ap.(\bar{f} \times id) = \bar{f} \stackrel{73, 73}{=} ap.(\hat{g} \times id) = \hat{g} \stackrel{73, 73}{=} ap(g \times id)(a, b) = \hat{g}(a, b) \stackrel{78}{=} ap(g a, id b) = \hat{g}(a, b)$$

$$\stackrel{74}{=} ap(g a, b) = \hat{g}(a, b) \stackrel{84}{=} g \ a \ b = \hat{g}(a, b) \stackrel{86}{=} g \ a \ b = \text{uncurry } g(a, b) \quad c.q.m.$$

$$6. \quad \overline{f \cdot (g \times h)} = \overline{ap.(\overline{id \times h}) \cdot \bar{f} \cdot g} \stackrel{25}{=} \overline{f \cdot (g \times h)} = ap.(\overline{ap.(\overline{id \times h}) \cdot \bar{f} \cdot g} \times id) \equiv$$

$$\stackrel{38, 1}{=} \overline{f \cdot (g \times h)} = ap.(\overline{(ap.(\overline{id \times h}) \cdot \bar{f} \cdot (g \times id)) \times id \cdot id}) \equiv \overline{f \cdot (g \times h)} = ap.(\overline{(ap.(\overline{id \times h}) \times id) \cdot (\bar{f} \cdot (g \times id)) \times id}) \equiv$$

$$\stackrel{36}{=} ap.(\overline{(id \times h) \cdot (\bar{f} \cdot (g \times id) \times id)}) \stackrel{14}{=} ap.(\overline{id \cdot (\bar{f} \cdot (g \times id)) \times (h \cdot id)}) \stackrel{1}{=} ap.(\overline{\bar{f} \cdot (g \times id) \times (id, h)})$$

$$\stackrel{38}{=} ap.(\overline{(\bar{f} \cdot g) \times (id, h)}) \stackrel{14}{=} ap.(\overline{(\bar{f} \times id) \cdot (g \times h)}) \stackrel{2}{=} (ap.(\bar{f} \times id)) \cdot (g \times h) \stackrel{36}{=} \bar{f} \cdot (g \times h) \quad c.q.m.$$

$$7. \quad \text{flip } f = \bar{f} \cdot \text{swap}$$

$$f \circ (f \circ f) \circ f = f \circ (f \circ \text{swap}) \circ f = f \circ \text{swap} \circ f = f \circ \text{swap} \circ \text{swap} = f \circ \text{id} = f = f \text{ c.q.m.}$$

$$\text{flip } f * y = f y * \equiv \widehat{f} \cdot \text{swap} * y \equiv \widehat{f} y * \equiv f y * \text{ c.q.m.}$$

8. $\text{junc} \cdot \text{unijunc} = \text{id}$
 $\text{unijunc} = \text{id}$
 $\text{junc}(f, g) = [f, g]$
 $\text{unijunc } K = (K \cdot i_1, K \cdot i_2)$

$$\text{junc} \cdot \text{unijunc} = \text{id} \stackrel{12, 23, 34}{=} \text{junc}(\text{unijunc } K) = K \stackrel{\text{unijunc}}{=} \text{junc}(K \cdot i_1, K \cdot i_2) = K \stackrel{12}{=} [K \cdot i_1, K \cdot i_2] = K \stackrel{20}{=} K \cdot [i_1, i_2] = K \stackrel{19, 1}{=} K = K \text{ c.q.m.}$$

$$\text{unijunc} \cdot \text{junc} = \text{id} \stackrel{12, 23, 34}{=} \text{unijunc}(\text{junc}(f, g)) = (f, g) \stackrel{\text{junc}}{=} \text{junc}[f, g] = (f, g) \stackrel{\text{unijunc}}{=} ([f, g] \cdot i_1, [f, g] \cdot i_2) = (f, g) \stackrel{12}{=} (f, g) = (f, g) \text{ c.q.m.}$$

9. $\text{in}: (A \times A) \times A \rightarrow \text{Point } A$

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$$(A \times B) \times C \rightarrow A \times (B \times C)$$

$$\beta: (A \times B) \times C \rightarrow A \times C$$

$$\begin{array}{ccc} (A \times B) \times C & \xrightarrow{\beta} & A \times C \\ \downarrow & & \downarrow \\ (f \times g) \times h & & f \times h \\ \downarrow & & \downarrow \\ (A' \times B') \times C' & \xrightarrow{\beta} & A' \times C' \end{array}$$

$$\beta((f \times g) \times h) = (f \times h) \cdot \beta$$

$$2. \pi_1 = \text{const} \stackrel{(75)}{=} \text{ap} \cdot (\text{const} \times \text{id}) = \pi_1 \stackrel{(72)}{=} \text{ap} \cdot (\text{const} \times \text{id}) (a, b) = \pi_1 (a, b) \stackrel{(72, 75)}{=} \text{ap}(\text{const} \times \text{id})(a, b) = a =$$

$$\stackrel{(78)}{=} \text{ap}(\text{const } a, \text{id } b) = a \stackrel{(6, 74)}{=} \text{ap}(\underline{a}, b) = a \stackrel{(84)}{=} \underline{a} \cdot b = a \stackrel{(75)}{=} a = a \quad \text{c.g.m.}$$

$$\pi_2 = \text{id} \stackrel{(35)}{=} \pi_2 = \text{ap} \cdot (\text{id} \times \text{id}) \stackrel{(72)}{=} \text{ap} \cdot (\text{id} \times \text{id}) (a, b) = \pi_2 (a, b) \stackrel{(72, 75, 78)}{=} \text{ap}(\text{id } a, \text{id } b) = b \stackrel{(74, 75)}{=} \text{ap}(\text{id}, b) = b =$$

$$\stackrel{(74)}{=} \text{id } b = b \stackrel{(74)}{=} b = b \quad \text{c.g.m.}$$

$$3. \text{List}([c], g, x)$$

$$g_x(a, b : x) = b : a : x$$

$$\begin{array}{ccc} \text{List}(A) & \xrightarrow{(c:)} & \text{List}(B) \\ \downarrow & & \uparrow \\ 1 + (A \times \text{List}(A)) & \longrightarrow & 1 + (B \times \text{List}(B)) \end{array}$$

$$4. \langle f, g \rangle = \text{for loop}(0, 0) \text{ where} \\ \text{loop}(a, b) = (a + b, h \ b) \\ h \ b = b + \text{ if even } b \text{ then } 1 \text{ else } (-1)$$

$$\begin{array}{l} Ff = \text{id} + f \\ \text{out}_{v_0} = \text{in} \\ \text{in} = [0, \text{succ}] \end{array}$$

$$\left\{ \begin{array}{l} f = [0, \text{add}] \cdot F \langle f, g \rangle \cdot \text{out}_{v_0}^{F_{\text{out}}} \\ g = [0, h] \cdot F g \cdot \text{out}_{v_0} \end{array} \right\} \equiv \left\{ \begin{array}{l} f = [0, \text{add}] \cdot (\text{id} + \langle f, g \rangle) \cdot \text{in}^0 \\ g = [0, h] \cdot (\text{id} + g) \cdot \text{in}^0 \end{array} \right\}$$

$$\equiv \left\{ \begin{array}{l} f \cdot \text{in} = [0, \text{add}] \cdot (\text{id} + \langle f, g \rangle) \\ g \cdot \text{in} = [0, h] \cdot (\text{id} + g) \end{array} \right\} \equiv \left\{ \begin{array}{l} f \cdot \text{in} = [0, \text{add}] \cdot (\text{id} + \langle f, g \rangle) \\ g \cdot \text{in} = [0, h] \cdot (\text{id} + (\pi_a \cdot \langle f, g \rangle)) \end{array} \right\}$$

$$\text{filter } p \cdot \text{filter } p = \text{filter } p \equiv \text{filter } p \cdot \text{concat} \cdot \text{map}(p \rightarrow \text{singl}, \text{nil}) = \text{filter } p \equiv$$

$$\equiv \text{concat} \cdot \text{map}(\text{filter } p) \cdot \text{map}(p \rightarrow \text{singl}, \text{nil}) = \text{filter } p \equiv \text{concat} \cdot \text{map}(\text{filter } p \cdot (p \rightarrow \text{singl}, \text{nil})) = \text{filter } p \equiv$$

$$\equiv \text{concat} \cdot \text{map}(p \rightarrow \text{singl}, \text{nil}) = \text{filter } p$$

